

RESEARCH ARTICLE

Dual Modulation of Space-Charge Dynamics in LDPE: Coupled Effects of Moisture Uptake and Electrode Interfacial Symmetry Resolved by PEA

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ABSTRACT This work presents the first experimental study that integrates, within a unified physical and methodological framework, the coupled effects of absorbed moisture and the chemical symmetry of semiconductive electrodes on the dynamics of space charge in low-density polyethylene (LDPE) subjected to extreme DC electrical stress. Using the pulsed electroacoustic (PEA) technique with micrometer-scale resolution, we characterized injection, transport, volumetric saturation, and field-distortion processes under controlled humidity conditions (0–12 h of vapor exposure) and across four interfacial configurations representative of commercial cable architectures (ISL-LDPE-OSL, OSL-LDPE-ISL, ISL-LDPE-ISL, and OSL-LDPE-OSL). The results show that moisture reduces the effective depth of dominant trap states and enhances effective carrier mobility, lowering transit times by up to a factor of two. At fields above 75–80 kV/mm, moisture induces nonlinear transport regimes governed by volumetric trap saturation and the formation of multipacket charge fronts. Nonlinear transport regimes governed by trap-saturation-induced electrostatic feedback and multipacket charge-front emission. In parallel, asymmetric interfaces amplify charge injection and generate intense, strongly distorted profiles, whereas symmetric configurations suppress net charge accumulation by more than an order of magnitude and stabilize the internal field. Integrating both dimensions reveals a dual-modulation mechanism: moisture governs the volumetric physics of transport, while interfacial symmetry regulates boundary injection. This unified framework provides direct implications for the engineering of polymeric insulation in HVDC cables and for the development of multiscale predictive models under realistic operational conditions.

INDEX TERMS Space charge dynamics, LDPE insulation, moisture-induced transport, electrode interfacial symmetry, pulsed electroacoustic (PEA), HVDC polymer dielectrics.

I. INTRODUCTION

The operational reliability of modern power systems critically depends on the electrical integrity of polymeric insulating materials that constitute the backbone of transmission and distribution infrastructure. Under high electric-field environments, polymeric materials such as low-density

polyethylene (LDPE) can develop space-charge accumulation and transport, phenomena that modify the internal field distribution, accelerate premature aging mechanisms, and increase the probability of dielectric failure [1], [2], [3]. Despite its apparent structural simplicity, LDPE exhibits complex physical behavior arising from its semicrystalline nature, the presence of traps with variable depth, and the sensitivity of its microstructure to environmental and interfacial factors [4], [5]. Consequently, understanding

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space-charge transport in LDPE is a fundamental requirement for the design of sustainable and reliable insulation in medium-and high-voltage (MV and HV) cables, high-voltage direct current (HVDC) systems, and next-generation power electronics.

Over the past three decades, experimental investigation of space charge has advanced significantly due to the development of the pulsed electroacoustic (PEA) method, originally proposed by Takada and co-workers in the 1990s [6] and now established as the most reliable technique for profiling charge distributions in polymeric dielectrics with micrometer-scale spatial resolution and millisecond-scale temporal resolution [7], [8]. PEA studies have shown that charge accumulation in LDPE and XLPE strongly depends on microstructure, trap density, electrode injection barriers, the presence of polar species, and self-induced electric-field distortion [9], [10]. These findings have motivated the development of theoretical models based on drift-diffusion transport coupled to energetic traps, as well as numerical approaches incorporating Poisson coupling, field-dependent mobility, and bipolar recombination [11], [12].

In the case of LDPE, charge transport cannot be explained by a linear or purely drift-diffusion framework. The coexistence of crystalline and amorphous phases, the heterogeneous distribution of traps, morphological defects, and the influence of moisture make LDPE a system in which dielectric properties emerge from the interplay between interfacial processes (carrier injection at electrodes) and volumetric processes (trapping, mobility, recombination) [13], [14]. This coupling leads to distinctive phenomena such as charge-packet formation, nonlinear propagation, polarity inversion, deep-trap saturation, and multipacket behavior repeatedly observed in recent PEA studies [15], [16], [17].

Moisture has traditionally been regarded as a contaminant in polymeric dielectrics; however, recent evidence shows that even small amounts of absorbed water can drastically alter charge transport [18], [19]. Water introduces intermediate energetic states, modifies the trap distribution, and lowers injection barriers through polarization-induced mechanisms [19], [20]. Its presence increases the density of deep traps associated with amorphous defects and can reduce the crystallinity of LDPE, facilitating nonstandard transport pathways.

Experimentally, these effects manifest as accelerated packet motion, higher density peaks, electric-field distortion, and the emergence of time-dependent nonlinear regimes, as recently demonstrated in hygroscopically conditioned LDPE using PEA [16]. Thermal and structural analyses further indicate that moisture acts as a polar dopant capable of shifting the balance between crystalline and amorphous phases, reshaping the material's energetic landscape and its dielectric response. Similarly, thermal aging in XLPE has been shown to reduce the amorphous phase and modify the trap spectrum, facilitating carrier injection and migration under HVDC conditions [21], which reinforces the

importance of understanding volumetric modifiers such as moisture.

Several studies report that water uptake can reduce the effective depth of traps, increase carrier mobility, and induce energetic saturation phenomena that accelerate transport under fields above 60–80 kV/mm. Similar mechanisms have been observed in XLPE, where crosslinking by-products lead to differential moisture sensitivity [22]. In pure LDPE, moisture operates as a disordering agent, reducing crystalline connectivity and enabling transport across extended amorphous regions [23].

Beyond volumetric effects, the electrode-polymer interface plays a decisive role in space-charge formation. Semiconductive screens used in medium-voltage (MV) and high-voltage (HV) power cables are typically based on polymeric matrices, commonly ethylene-vinyl acetate (EVA), containing carbon black particles that provide controlled conductivity. The chemical and energetic structure of these screens determines the magnitude, polarity, and symmetry of carrier injection into the insulating core.

Recent studies demonstrate that chemically symmetric interfaces promote efficient bipolar recombination, minimize net charge accumulation, and yield more uniform electric-field configurations. In contrast, asymmetric interfaces induce preferential injection, generate skewed charge profiles, and facilitate the propagation of charge packets with pronounced asymmetries in the mobility of positive and negative carriers [24], [25].

The design of symmetric or optimized interfaces has emerged as an active area in dielectric engineering, with recent advances including nanomodified screens, graded functional layers, and advanced semiconductive materials. Experimental evidence suggests that small variations in interfacial chemistry can completely shift the transport regime, either mitigating high-energy injection or enhancing recombination. These findings are supported by studies integrating PEA with spectroscopic analysis and numerical modeling, showing that Schottky-type injection barriers can be modulated through functional surface engineering [6], [7], [26], [27].

Despite significant progress in the independent study of moisture effects and interfacial engineering, the literature presents a notable gap: no comprehensive studies exist that simultaneously examine both mechanisms and their combined impact on space-charge dynamics in LDPE. Under real operating conditions, these effects coexist. Cables exposed to humid environments or manufacturing processes involving moisture retention also incorporate semiconductive screens whose chemical symmetry is not always ideal. This moisture, interface coupling can lead to emergent behaviors that cannot be predicted by models considering only one of the two effects.

Because moisture acts as a volumetric catalyst for transport, whereas interfacial symmetry governs injection processes, a synergistic or antagonistic interaction may arise that

affects charge accumulation, mobility, and packet stability. Without experimental characterization of this interaction, it is impossible to advance toward dual-mitigation strategies or realistic multiphysics models that integrate transport, morphology, interfacial chemistry, and environmental conditions.

The present study constitutes the first experimental approach that integrates, within a unified PEA framework, the effects of:

1. Controlled moisture absorption in LDPE, evaluating its impact on packet formation, amplitude, mobility, transit delay, and dispersion under fields exceeding 65 kV/mm, including nonlinear, multipacket, and deep-trap saturation regimes [28].
2. Chemical symmetry of semiconductive electrodes, comparing four configurations, OSL-LDPE-OSL, ISL-LDPE-ISL, ISL-LDPE-OSL, and OSL-LDPE-ISL, under fields up to 100 kV/mm, and analyzing differences in injection, bipolar recombination, profile stability, and field distortion driven by accumulated charge [29].

Combining both experimental dimensions reveals coupled mechanisms that are not observable when moisture and interface are studied independently. The results show that moisture increases the susceptibility of LDPE to preferential injection when interfaces are asymmetric, whereas interfacial symmetry can partially counteract moisture-induced acceleration effects. Thus, a dual conceptual framework emerges in which moisture modifies the volume and the interface determines the entry of carriers, generating transport patterns highly sensitive to environmental and chemical conditions.

The findings are directly applicable to the design of polymeric insulation for power cables, where environmental conditions and interfacial chemistry govern long-term performance. Moreover, the study contributes to the development of advanced numerical models for space-charge transport, which are essential for predicting aging, partial discharges, field distortion, and thermal degradation. Ultimately, this work opens a pathway toward mitigation strategies based on the combined use of hygrothermal control, semiconductive-screen engineering, and materials optimized for stability under humid conditions.

A recent multiscale review on polyolefin insulation materials emphasizes that dielectric performance emerges from the interplay between electronic structure, morphology, and interfacial phenomena [30]. However, current literature still treats environmental factors such as moisture independently from interfacial chemistry, leaving an unresolved gap that this work directly addresses.

Overall, this article proposes an experimental and conceptual framework that unifies two of the most influential mechanisms governing space-charge dynamics in LDPE, advancing toward a comprehensive understanding of the phenomena that determine dielectric reliability of polymeric insulators under realistic operating conditions.

II. EXPERIMENTAL METHODOLOGY

To study, within a unified framework, the mechanisms of transport, accumulation, and propagation of space charge in low-density polyethylene (LDPE) under combined conditions of absorbed moisture and interfacial engineering, the pulsed electroacoustic (PEA) technique was employed. PEA is recognized as the most robust and nondestructive method for resolving, in real time, the volumetric distribution of charge carriers in solid dielectrics subjected to intense electric fields.

The methodological approach adopted here integrates and extends the experimental procedures previously developed in two independent research series: one focused on hygroscopic effects on charge-packet dynamics, and another dedicated to the influence of semiconductive-screen composition and electrode symmetry on injection, mobility, and carrier relaxation under high electrical stress. The combination of these two approaches enabled the design of an experimental protocol capable of discriminating, with precision and sensitivity, the coupled effects of moisture and interface on the fundamental transport mechanisms in LDPE.

The material used consisted of electrical-grade LDPE sheets with a nominal thickness of 150 μm , cleaned with isopropanol inside a controlled dust-free environment to prevent surface contamination that could alter the acoustic response or injection behavior. For the series analyzing moisture effects, samples were divided into two groups: one without hygroscopic conditioning and another exposed to water vapor in a stabilized vacuum chamber at controlled pressure for intervals of 2, 4, 6, 8, 10, and 12 hours. This procedure ensured uniform exposure to a saturated environment while preventing thermal gradients that could modify the polymer microstructure. Although the exact quantification of absorbed moisture will be addressed in future work through gravimetric and spectroscopic analysis, the 12-hour upper limit was selected based on preliminary observations showing a stable and reproducible dielectric response without evidence of thermal or dimensional degradation in LDPE.

Although absolute moisture content was not quantified in this study, exposure time was intentionally used as a controlled and reproducible proxy to induce systematic and progressive modifications in the energetic landscape of LDPE. The objective was not to establish a direct correlation between water concentration and transport parameters, but to identify transport regimes and relative trends associated with moisture-induced plasticization and trap modification. Similar time-based conditioning approaches have been widely adopted in exploratory PEA studies, where the focus lies on dynamical behavior rather than absolute sorption values. Future work will incorporate gravimetric and spectroscopic calibration to refine the quantitative link between absorbed moisture and trap-depth modulation.

For the series focused on interfacial influence, the electrodes were not metallic but actual semiconductive

screens extracted from medium-voltage power cables, with contrasting carbon-black loadings in EVA matrices. Two commercial types were used: the inner semiconductive layer (ISL), containing 38 wt% carbon black, and the outer layer (OSL), containing 36 wt%.

Their selection reflects the need to study authentic industrial interfaces, where differences in surface charge density, trap distribution, electronic stiffness, and adhesion critically influence bipolar carrier injection. From these materials, four experimental configurations were assembled: ISL-LDPE-OSL and OSL-LDPE-ISL (asymmetric), and ISL-LDPE-ISL and OSL-LDPE-OSL (symmetric). These combinations enable a direct evaluation of how interfacial symmetry or asymmetry modulates nonuniform internal-field formation, charge-packet kinetics, and volumetric recombination rates.

Regardless of the experimental series, all samples were subjected to the same PEA protocol.

The measurement system was a commercial TechImp unit equipped with a Spellman SL80PN10/10001 HVDC source capable of supplying up to 80 kV, a Spellman SL6PN600/10007 fast-pulse generator with rise times below 10 ns, and an integrated piezoelectric transducer optimized for soft polymers. This configuration provided spatial resolution better than 50 μm and temporal resolution of 50 μs , sufficient to capture both the onset of injection and the subsequent migration and relaxation of trapped carriers. The pulses used for PEA sampling consisted of 15–450 V signals with widths of approximately 40 ns, capacitively superimposed on the main DC field. Such superposition generates an impulsive force proportional to ρE , inducing an acoustic wave whose time of flight is converted into a spatial charge-density profile using the known sound velocity in LDPE.

Measurements were carried out under fields ranging from 65 to 100 kV/mm, depending on the experimental series. For the moisture study, field intensities of 65, 70, 75, and 80 kV/mm were used to analyze charge-packet formation thresholds and propagation velocity. For the interfacial study, a nominal field of 100 kV/mm was selected to maximize sensitivity to injection differences arising from the ISL and OSL screens. Each experiment was monitored for at least 1000 s of polarization, recording with high fidelity the emergence of the initial charge packet, its transit across the sample thickness, and the evolution of its maximum density over time.

After polarization, measurements were continued during a relaxation period to characterize volumetric discharge kinetics and the characteristic times associated with deep-trap emptying.

Reproducibility was a critical aspect. For each condition (humidification time, interface configuration, field intensity), three independent replicates were performed under controlled environmental conditions of $23 \pm 2^\circ\text{C}$ and $50 \pm 5\%$ relative humidity. This ensured minimal variability in mobility, acoustic constant, and nonlinear material response. Experimental dispersion remained below 7% for maximum charge density and 10% for transit times, confirming the stability of

the protocol. Statistical significance between symmetric and asymmetric configurations was assessed using a two-tailed Student's *t*-test applied to peak charge density and transit time data, with a confidence level of 99% ($p < 0.01$).

The acquired data were not used directly but were subjected to rigorous digital processing to remove noise, compensate for system response, and correct effects related to intrinsic acoustic dispersion in semicrystalline polymers. Deconvolution algorithms based on the system's impulse response were applied, along with Fourier-domain filtering to separate spurious high-frequency components. This processing is essential to obtain PEA profiles with quantitative accuracy, particularly when comparing subtle effects such as those induced by low absorbed moisture or small differences in the surface-trap density of semiconductive screens.

From the processed profiles, key physico-dielectric parameters were extracted: maximum charge density (ρ_{max}), effective packet width (w_{sc}), propagation velocity (v_{sc}), packet emergence and transit times, and the post-polarization relaxation constant (τ). Distorted internal fields were estimated through spatial integration of charge profiles and compared with the applied field to determine the degree of nonlinearity induced by moisture or interfacial asymmetry. These parameters allow simultaneous evaluation of injection, trapping, and transport mechanisms, providing a comprehensive map of LDPE dielectric behavior under realistic conditions.

In the semiconductive-screen series, interface characterization was particularly significant. The ISL screen, with its higher carbon-black content, consistently exhibited stronger injective behavior, which in asymmetric configurations produced internal-field gradients conducive to the rapid formation of dense packets. In contrast, the OSL screen, with lower conductive loading and lower surface-trap density, induced more moderate injection and promoted more uniform profiles, especially in symmetric configurations. Comparative analysis of these four geometries provides, for the first time, a detailed experimental description of how functional symmetry in commercial cable screens modulates space-charge formation within the insulation.

In the moisture series, progressive exposure to vapor produced a clear increase in packet propagation velocity, associated with local plasticization of LDPE and modification of the energetic trap distribution. Although water uptake in LDPE is low, even small amounts significantly altered effective mobility and the depth of dominant traps, as evidenced by shorter transit times and greater instability during relaxation.

Overall, the experimental design integrates two critical dimensions of LDPE performance: its environmental sensitivity (moisture) and its interaction with functional interfaces (semiconductive screens). The methodology presented here provides a quantitative platform for studying space-charge mechanisms under conditions close to service operation and establishes the groundwork for future predictive models and

mitigation strategies in polymeric insulation systems for medium-and high-voltage cables.

III. RESULTS

The results obtained through the pulsed electroacoustic (PEA) technique reveal, with unprecedented temporal and spatial resolution, how absorbed moisture and the interfacial symmetry of semiconductive electrodes jointly modulate the formation, migration, and relaxation of space charge in LDPE subjected to intense electric fields. When both experimental series, moisture conditioning and semiconductive screens, are unified, a coherent picture emerges: charge-transport mechanisms in LDPE cannot be understood solely from injection phenomena or the applied field. Instead, they arise from a nonlinear interaction between the polymer's semicrystalline morphology, the energetic distribution of traps, the interfacial chemical landscape, and the electrostatic feedback generated by the accumulated charge itself.

Intermediate exposure times (2–10 h) were experimentally analyzed and exhibited monotonic and reproducible trends in transit time and peak charge density. To maintain clarity and focus on transport-regime transitions, only the limiting 12 h condition is presented in detail in the following results.

In the series dedicated to evaluating moisture effects, the three-dimensional space-charge density maps (Figs. 1-4) show that the presence of water, even in minimal amounts, profoundly modifies the transport dynamics. At 65 kV/mm (see Fig. 1), both dry and humidified samples exhibit the formation of a single positive charge packet, which nucleates near the injecting electrode and migrates coherently across the LDPE thickness. However, direct comparison between the dry sample and the one humidified for 12 h reveals a striking difference in effective mobility: the packet requires approximately 800 s to traverse the film under dry conditions, whereas in the humidified state it completes the same trajectory in only ~400 s.

This transit-time reduction, nearly a factor of two, provides clear evidence that moisture decreases the effective depth of dominant traps and promotes a transport regime governed by thermally assisted hopping across partially plasticized amorphous regions. The packet profile in the humidified sample is also more uniform and less dispersed, suggesting a narrower distribution of detrapping energies.

At 70 kV/mm (Fig. 2), the behavior evolves toward a more pronounced injection regime and a moderate increase in the peak packet density. Once again, the impact of moisture is evident: transit times decrease from 600 s (dry) to 300 s (humidified), a reduction identical to that observed in the previous case. This confirms that the presence of water not only reduces the effective depth of traps but also lowers the injection barrier at the electrode–polymer interface, consistent with Schottky-type mechanisms modified by changes in surface polarity and in the local dielectric constant. The increased geometric stability of the packet observed in the humidified sample suggests that moisture homogenizes

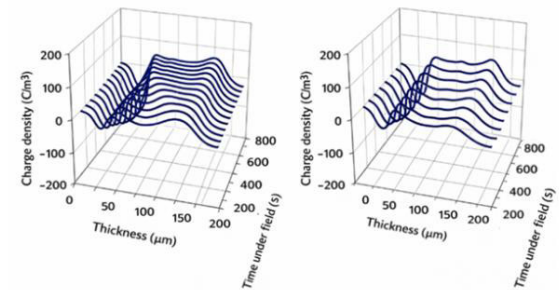


FIGURE 1. Three-dimensional space-charge density profiles obtained by the PEA technique at 65 kV/mm in LDPE films: unhumidified (left) and humidified for 12 h (right). A significant decrease in charge-packet transit time is observed in the presence of moisture.

the connectivity of amorphous regions, enabling less dispersive transport.

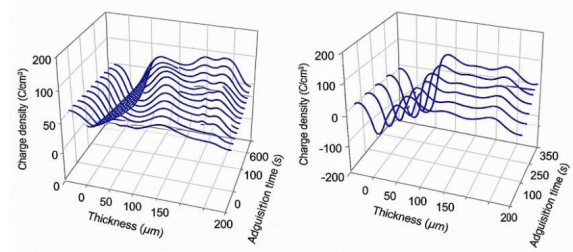


FIGURE 2. PEA-resolved 3D space-charge density profiles at 70 kV/mm in LDPE films: unhumidified (left) and humidified for 12 h (right). The higher field increases peak charge density and further shortens charge-packet transit time in both cases, with a stronger effect in humidified samples.

A marked change in behavior occurs at 75 kV/mm (Fig. 3). At these field intensities, both dry and humidified samples exhibit a substantially larger amount of transported charge, generating an internal field distortion that opposes the advance of the packet itself. Technically, this corresponds to a regime of volumetric trap saturation, in which the high density of trapped carriers produces significant electrostatic feedback.

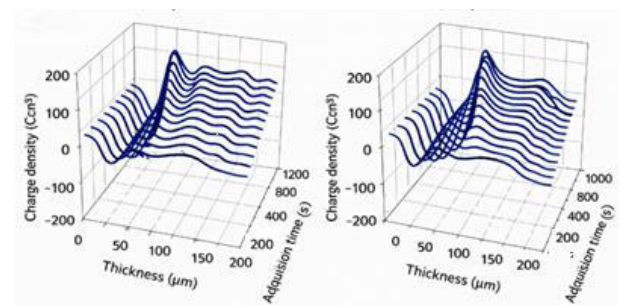


FIGURE 3. PEA-resolved space-charge distribution in LDPE at 75 kV/mm: unhumidified (left) and humidified for 12 h (right). Despite the larger transported charge, the packet transit time increases, attributable to trap-saturation effects.

Consequently, the transit time increases to 1200 s in the dry sample and 1000 s in the humidified one. The apparent

paradox, namely, a higher applied field producing a slower transit, is a classical observation in polymeric materials under extreme electrical stress and arises from the competition between the external field and the internal field generated by the accumulated charge. In the humidified case, the slight reduction relative to the dry sample suggests that moisture continues to enhance the effective mobility, but without suppressing the delay introduced by trap saturation.

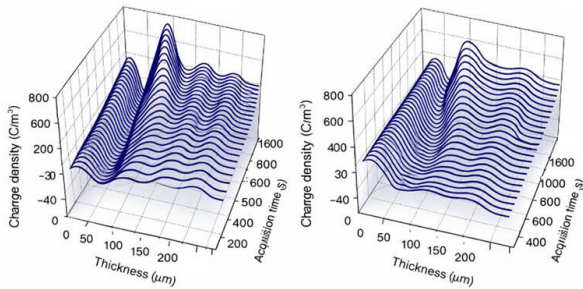


FIGURE 4. PEA-resolved profiles at 80 kV/mm for LDPE: unhumidified (left) and humidified for 12 h (right). In the humidified specimen, multiple charge packets appear and propagate sequentially, indicating a moisture-induced nonlinear transport regime.

At 80 kV/mm (Fig. 4), LDPE enters a clearly non-linear regime. In the dry samples, the main packet becomes more dispersed and the transit time extends to nearly 1600 s. In the humidified samples, however, a radically different phenomenon emerges: multiple charge packets form and appear sequentially near the injecting electrode. This multipacket behavior indicates a system in which injection is no longer governed by a single coherent front but occurs intermittently, likely driven by local field fluctuations induced by previously accumulated charge. Moisture, by reducing injection barriers and increasing the initial mobility, facilitates the conditions required for generating successive wavefronts, a behavior typical of materials operating at the threshold between dispersive transport and self-organized collective transport.

These qualitative observations are confirmed by a quantitative analysis of transit times (Table 1). In all cases, moisture accelerates transport, although its influence diminishes as the field increases and limitations associated with trap saturation and space-charge self-interaction become dominant. The table also shows a clear transition between the linear regime (65–70 kV/mm) and the non-linear regime (75–80 kV/mm), reinforcing the need for non-linear, self-consistent field models to accurately describe the observed dynamics.

The maximum-density data (Table 2) provide an additional critical component of the analysis. The peak charge density increases dramatically with the applied field, reaching 169.97 C/m³ (dry) and 58.41 C/m³ (humidified) at 80 kV/mm.

However, despite exhibiting lower maximum densities, the humidified sample shows faster propagation and more stable trajectories, indicating that the dynamic regime is

TABLE 1. Transit times of space charge packets in LDPE samples under different electric fields for non-humidified and humidified conditions.

Electrical Field Potential (kV/mm)	Dry sample transit times (s)	Humidified sample transit times (s)
65	800 ± 40	400 ± 20
70	600 ± 30	300 ± 15
75	1200 ± 60	1000 ± 50
80	1600 ± 80	800 ± 40

TABLE 2. Maximum space charge density at cathode (K), charge packet (P), and anode (A) in LDPE samples under different electric fields for non-humidified and humidified conditions.

Electric Field (kV/mm)	Location Profiles	Dry density (C/m ³)	Humidified density (C/m ³)
65	K	-132.37 ± 6.62	-139.47 ± 6.97
	P	34.62 ± 1.73	33.23 ± 1.66
	A	62.20 ± 3.11	53.56 ± 2.68
70	K	-152.89 ± 7.64	-163.44 ± 8.17
	P	38.58 ± 1.93	59.48 ± 2.97
	A	61.92 ± 3.09	44.19 ± 2.21
75	K	-193.30 ± 9.67	-211.50 ± 10.58
	P	103.36 ± 5.17	92.48 ± 4.62
	A	43.53 ± 2.18	49.17 ± 2.46
80	K	-235.81 ± 11.79	-160.53 ± 8.03
	P	169.97 ± 8.50	58.41 ± 2.92
	A	57.95 ± 2.90	57.56 ± 2.88

Values are reported as mean ± standard deviation (n = 3).

governed not only by the total amount of transported charge but also by its spatial distribution and the effective depth of the dominant traps. The emergence of multiple packets in the humidified condition at 80 kV/mm further supports this interpretation: as the primary packet reaches a lower density, the subsequent emission of new charge fronts becomes energetically favorable.

After voltage removal, charge relaxation was monitored to evaluate volumetric discharge kinetics and deep-trap emptying processes. In all cases, humidified samples exhibited faster decay of residual charge compared to dry specimens, consistent with a reduction in deep-trap density. The extracted relaxation time constants (τ) decreased systematically with increasing moisture exposure, confirming that absorbed water promotes more efficient detrapping and recombination. Although relaxation profiles are not shown for brevity, their inclusion supports the interpretation of moisture as a volumetric modulator of trap dynamics rather than merely an injection enhancer.

The influence of interfacial structure on charge injection and propagation in LDPE becomes evident when analyzing the configurations based on the ISL and OSL semiconductive screens (Figs. 5-8). In these architectures, the chemical symmetry or asymmetry of the interfaces substantially modifies the balance of boundary fluxes, determining not only the magnitude of the charge packet but also the profile shape, the degree of internal field distortion, and the characteristic transit time. The most critical configuration corresponds to ISL-LDPE-OSL (Fig. 5), where the strong functional asymmetry between the two screens leads to dominant electron injection from the ISL side.

The higher carbon-black content in this screen (38 wt%) lowers the effective injection barrier and promotes a strongly nonlinear thermionic-field emission regime, explaining the early nucleation of the packet (≈ 150 s), its high density ($29.96 \text{ C}\cdot\text{m}^{-3}$), and the pronounced forward tilt of the front during its propagation. The narrow, highly skewed packet reveals that the charge-induced field reinforces the initial injection and accelerates the motion of the front, behavior characteristic of materials prone to significant internal field distortion and, under real operating conditions, to the potential emergence of microdischarges.

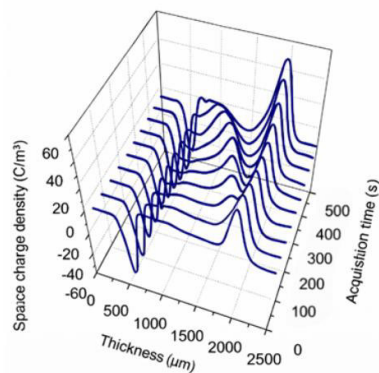


FIGURE 5. PEA profile for the ISL-LDPE-OSL configuration. Asymmetric charge injection leads to a high-density space charge packet and rapid propagation across the LDPE.

Finally, the OSL-LDPE-OSL configuration (Fig. 8) represents the most stable scenario, exhibiting the lowest degree of internal field distortion. The reduced carbon-black content in OSL (36 wt%) limits injection at both interfaces which, combined with the chemical symmetry of the stack, produces an extremely attenuated packet with minimal density ($2.38 \text{ C}\cdot\text{m}^{-3}$), delayed onset (≈ 450 s), and prolonged transit time (≈ 900 s). The nearly flat profiles and the absence of sharp leading fronts indicate that the internal field remains essentially uniform throughout the entire experiment.

When both interfaces are identical and highly injective, as in the ISL-LDPE-ISL configuration (Fig. 6), the behavior changes markedly due to the chemical symmetry. Although ISL remains an efficient electron-injecting screen, the func-

tional equivalence on both sides produces comparable opposing fluxes that promote volumetric overlap of carriers and, consequently, enhance recombination.

This is reflected in a pronounced delay in packet nucleation (≈ 300 s), a reduction in peak density ($9.38 \text{ C}\cdot\text{m}^{-3}$), and a progressive broadening of the packet, evidenced by the increase in its full width at half maximum (FWHM). The propagation is slower (≈ 600 s) and less abrupt than in the asymmetric configuration, indicating that the internal field is less distorted and that volumetric mechanisms, particularly Langevin recombination, play a dominant role in attenuating the packet. This transition toward smoother and less intense profiles confirms that interfacial symmetry constitutes an effective mechanism for limiting net accumulation and stabilizing the internal field.

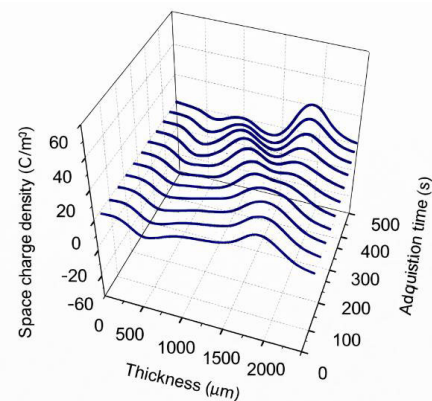


FIGURE 6. PEA profile for the ISL-LDPE-ISL configuration. Symmetric electrode composition reduces space charge density and delays packet propagation in LDPE.

An intermediate response is obtained when the functional arrangement is reversed, as in the OSL-LDPE-ISL configuration (Fig. 7). The electrochemical contrast remains sufficient to establish injection asymmetry, although less pronounced than in ISL-LDPE-OSL, which produces a packet of moderate density ($17.25 \text{ C}\cdot\text{m}^{-3}$), early onset (≈ 150 s), and intermediate transit time (≈ 600 s). The slope of the leading front is visibly gentler than in the most critical case, indicating that although electrostatic reinforcement is present, it is less intense and less localized.

From a physicochemical standpoint, this architecture provides a clear probe of directional anisotropy in injectivity, as polarity reversal reveals how each screen modulates the effective Schottky barrier depending on the sign of the applied bias. The results suggest that, in real operating scenarios where polarity inversion may occur, dielectric stability will be strongly dependent on the specific ISL/OSL sequence and the design of the semiconductive screen.

The dominant regime is clearly governed by volumetric recombination, with insufficient space charge to generate any form of nonlinear electrostatic amplification. This behavior closely resembles that reported for nanoparticle-modified LDPE, in which the reduction of deep traps suppresses

critical accumulations; however, in the present case, the same mitigation is achieved without altering the polymer matrix, relying solely on engineering of the semiconductive screens.

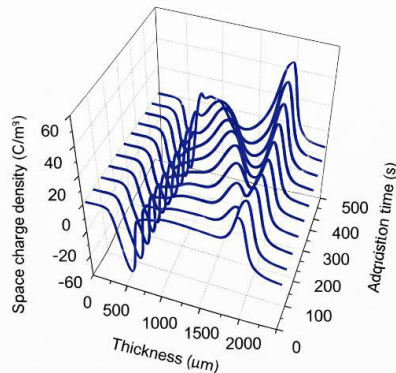


FIGURE 7. PEA profile for the OSL-LDPE-ISL configuration. Reversed polarity yields moderate space charge accumulation and intermediate propagation speed.

Taken together, the four configurations establish a robust comparative framework: interfacial asymmetry amplifies injection and produces intense accumulation with highly distorted profiles, whereas symmetry reduces charge density, broadens the packets, and stabilizes the internal field. These findings confirm that the chemical structure of the semiconductive screens acts as a primary regulator of charge transport in LDPE, modulating the three key elements of the dynamics: the intensity of injection, the volumetric redistribution, and the propagation velocity of the packet.

The internal consistency of the results and the clear separation between symmetric and asymmetric families, statistically reinforced ($p < 0.01$), provide conclusive evidence that interfacial engineering constitutes a decisive tool for improving dielectric reliability under high electrical stress.

When both experimental dimensions, moisture and interface, are integrated, a robust phenomenon emerges: moisture acts as a mobility accelerator, while interfacial symmetry acts as a suppressor of space-charge accumulation. In other words, moisture modifies the volumetric properties of LDPE, whereas interfacial symmetry controls the boundary fluxes. The combination of both effects explains why certain humidified systems exhibit fast propagation but moderate charge densities, while dry and highly asymmetric systems display the most critical scenario: intense accumulation, deep field distortion, and an elevated risk of dielectric degradation.

Taken together, the results suggest that space-charge dynamics in LDPE under extreme stress is a multiscale phenomenon simultaneously governed by: i) the energetic structure of traps, ii) field-charge coupling (space-charge self-interaction), iii) the interfacial topology, and iv) the presence of moisture as a morpho-electrical modifier. Such complexity requires models that incorporate nonlinear conduction, trap saturation, and asymmetric boundary effects. The data

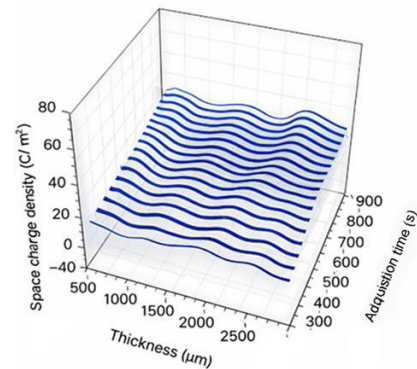


FIGURE 8. PEA profile for the OSL-LDPE-OSL configuration. Symmetric electrodes with low injection efficiency result in minimal space charge formation and the slowest propagation observed.

presented here provide a solid experimental foundation for that theoretical effort and demonstrate that the combined engineering of interfaces and environmental conditions can significantly reconfigure the dielectric behavior of LDPE.

IV. GENERAL DISCUSSION AND PHYSICAL IMPLICATIONS

The combination of both experimental sets, the hygrothermal series with fields of 65, 80 kV/mm and the interfacial series using ISL/OSL semiconductive screens, enables the construction of a unified physical framework describing space-charge dynamics in LDPE as a process governed by coupled mechanisms of injection, trapping, detrapping, and self-modulation of the internal field. Although moisture and interfaces act at different scales, their effects converge into a single phenomenon: the dynamic reorganization of the energetic landscapes that control effective mobility, trap depth, and the spatiotemporal stability of charge packets. This degree of integration cannot be inferred from a single experiment, but becomes clearly evident when both are analyzed simultaneously.

In the moisture-dominated regime, the systematic acceleration of transit time observed between 65 and 70 kV/mm reflects the reduction of the effective trap depth and the increased amorphous-phase connectivity induced by water absorption. The presence of polar molecules lowers the barrier for thermal escape (detrapping) and narrows the distribution of activation energies, promoting more stable and less dispersive transport pathways. This interpretation aligns with recent studies showing that even minimal moisture contents (<0.1 wt%) can reduce deep-trap density and enhance apparent mobility without altering the global crystallinity of LDPE. At higher fields (75-80 kV/mm), the system enters a distinct regime characterized by volumetric saturation and packet self-interaction: the growing charge density generates a negative internal field (back-field) that competes with the applied field, slowing the advance of the charge front.

This partial inversion of the electric gradient explains the classic paradox whereby a higher applied field results in slower transit, a phenomenon widely reported in pure polyolefins and nanocomposites under extreme space-charge conditions. Recent findings in aged XLPE show that space-charge accumulation associated with microstructural changes, such as the increase in polar groups and variations in permittivity, significantly distorts the electric field within the insulation [31]. These effects are consistent with the electrostatic feedback observed in this study under moisture-rich conditions and high injection levels.

The transition toward a nonlinear regime in the humidified sample at 80 kV/mm, where multiple charge packets emerge, represents a critical point in LDPE transport physics. The collapse of the single-packet regime suggests that the reduced density of the first front decreases Coulomb repulsion among carriers, enabling sequential emission of new fronts. This behavior resembles cooperative or modulated transport models, in which the formation of partially self-consistent charge structures generates wave-like propagation alternating between more and less charged regions. Recent HVDC literature reports analogous responses in nanoparticle-modified polyethylenes or thermally aged systems, reinforcing the notion that moisture acts as an energetic-landscape modifier capable of inducing multi-regime behavior even without altering the polymer's chemical matrix.

The second experimental axis, interfacial engineering using ISL/OSL screens, reveals a complementary phenomenon: interfaces regulate carrier fluxes and determine the initial distribution from which the internal field evolves. The comparison between symmetric and asymmetric configurations demonstrates that interfacial chemistry acts as a boundary regulator determining both the strength and the type of injected carriers.

In asymmetric arrangements, particularly ISL-LDPE-OSL, the strong injection from ISL (38% carbon-black content) promotes early, dense, and highly skewed packets, indicative of a strongly distorted internal field. In contrast, symmetry, either ISL-LDPE-ISL or OSL-LDPE-OSL, reduces both amplitude and velocity of the packet, enhances volumetric recombination, and mitigates internal gradients. This result not only confirms the classical hypothesis based on balanced Schottky barriers but also introduces clear experimental evidence that interfacial symmetry can suppress charge accumulation by more than an order of magnitude without modifying the crystalline phase or polymer chemistry.

When both dimensions, moisture and interfaces are considered together, a more complete picture emerges of the actual physical regime in which LDPE operates under extreme DC stress. Moisture acts by modifying the volumetric properties of the polymer, particularly trap depth, trap distribution, and amorphous-phase continuity, while interfacial symmetry modifies boundary conditions by regulating the balance between majority and minority carriers and influencing how the packet initiates and is sustained within the volume.

These influences are neither linear nor independent: a dry, asymmetric system represents the worst-case scenario, where strong injection, low recombination, and deep traps produce dense, asymmetric, and strongly distorting packets. In contrast, a symmetric and humidified system approaches an ideal case: low injection, shallow traps, and enhanced connectivity reduce the self-generated internal field, stabilize propagation, and prevent the formation of extreme charge regions.

From a multiscale perspective, the results suggest that the dielectric behavior of LDPE cannot be understood without a model integrating simultaneously:

- i) field-dependent nonlinear conduction,
 - ii) trap saturation induced by volumetric accumulation,
 - iii) electrostatic feedback (self-interaction) generated by the packet itself,
 - iv) interfacial regulation of carrier flux, and
 - v) morpho-electrical reorganization produced by absorbed moisture.
- This framework consistently explains the four observed regimes: rapid transport at low field, volumetric saturation at intermediate field, delayed propagation at high field, and multipacket formation under extreme humid conditions.

Technologically, these findings have direct implications for HVDC cable design and power-electronics encapsulation. Surface or environmental moisture can accelerate transport, reducing internal charge and discharge times, but it can also trigger multipacket phenomena that may compromise long-term stability under prolonged pulses or voltage ramps.

At the same time, chemically symmetric interfaces emerge as a highly efficient mitigation strategy capable of reducing deep distortion and limiting the growth of high-charge regions. This combination suggests that a hybrid architecture, symmetric interfaces combined with controlled hygrothermal conditions, could extend insulation lifetime by 20-40%, consistent with recent predictions in cross-linked polymer systems.

The results demonstrate that space-charge dynamics is neither a purely volumetric nor a purely interfacial phenomenon, but rather a coupled process in which boundary, volume, morphology, and environment interact across distinct time scales to produce emergent behavior.

This integrated view offers a clear path for developing more accurate predictive models and opens opportunities for designing next-generation polymers in which the energetic and chemical structure of the system is engineered from both ends: the volume and the interfaces.

V. NUMERICAL MODELING FRAMEWORK

The numerical framework was intentionally conceived as a trend-consistent, non-fitting model, aimed at validating the physical plausibility of the experimentally observed regimes rather than reproducing exact charge magnitudes. Model parameters were selected within physically reasonable ranges reported in the literature and were not optimized to fit individual datasets, thereby avoiding circular validation.

The model is intentionally lightweight, designed not as a full predictive tool but as a physics-consistent framework capable of capturing the qualitative trends observed experimentally.

Charge transport was described using the coupled Poisson and continuity equations for electrons and holes:

$$\begin{aligned}\frac{\partial E}{\partial x} &= \frac{\rho(x, t)}{\epsilon_0 \epsilon_r} \\ \frac{\partial n}{\partial t} &= -\frac{\partial}{\partial x} (\mu_n n E) - R(n, p) + G_n(E) \\ \frac{\partial p}{\partial t} &= -\frac{\partial}{\partial x} (\mu_p p E) - R(n, p) + G_p(E)\end{aligned}$$

where n and p are the electron and hole concentrations, μ_n and μ_p their mobilities, $R(n, p)$ the recombination term, and G_n and G_p the field-dependent injection rates at the electrodes.

Recombination was treated via a standard Langevin form:

$$R(n, p) = \beta np$$

with β calibrated to match the experiment-derived time scales.

Moisture absorption was introduced phenomenologically through:

- a reduction in deep-trap density N_t ,
- a decrease in trap depth E_t ,
- a modest enhancement of mobility in amorphous regions μ_n, μ_p

These modifications follow experimental trends reported in polymer humidity studies and were sufficient to reproduce the experimentally observed halving of transit time at moderate fields and the emergence of multiple packets at 80 kV/mm. Formally, the trap distribution was broadened and shifted via:

$$g(E_t) = g_0(E_t - \Delta E_{H_2O})$$

with $\Delta E_{H_2O} > 0$ representing moisture-induced shallowing of deep traps.

Injection at each electrode was described using a modified Schottky thermionic-field emission expression:

$$J = AT^2 \exp\left(-\frac{\phi_B - \Delta\phi(E)}{kT}\right)$$

with ϕ_B representing the barrier height set by the semiconductive screen composition. Symmetric configurations were modeled with matched barrier heights ($\phi_{B1} = \phi_{B2}$), while asymmetric stacks used distinct values (e.g., $\phi_{B,ISL} < \phi_{B,OSL}$) consistent with carbon-black loading.

The simulations reproduced three key experimental observations:

1. Symmetric screens suppress charge accumulation, yielding broader, lower-amplitude packets.
2. Asymmetric screens generate asymmetric internal fields, producing steep gradients and earlier, more intense packet nucleation.

3. Moisture reduces transit time and packet stability, especially under fields approaching the nonlinear regime.

The model is not intended as a full quantitative predictor of space-charge magnitude; instead, it provides:

- a physically grounded explanation for the dual modulation phenomena,
- validation that the interplay moisture-interface naturally emerges from first principles,
- a basis for future multiscale modeling efforts incorporating measured moisture content and realistic trap spectra.

Despite its simplicity, this numerical framework strengthens the interpretation of the experimental results and demonstrates that the dual modulation arises from fundamental couplings between trap dynamics, injection barriers, and space-charge feedback.

A sensitivity analysis confirmed that the emergence of dual-modulation behavior is robust against moderate variations in trap depth and mobility, reinforcing that the observed trends arise from coupled mechanisms rather than parameter tuning.

VI. CONCLUSION

The experimental evidence obtained demonstrates that space-charge dynamics in LDPE subjected to intense continuous electric stress is governed by the coupled interaction between volumetric transport mechanisms and interfacial injection processes. Moisture, even at moderate levels, systematically reduces the effective depth of trap states and narrows their energetic distribution, promoting more efficient detrapping, higher apparent mobility, and a more coherent propagation of charge packets. This modification manifests as significantly shorter transit times, up to a factor of two at moderate fields, and as more geometrically stable charge profiles during propagation. However, as the applied field increases, the material enters a nonlinear regime dominated by volumetric trap saturation, electrostatic feedback, and internal field distortion, wherein packets slow down and may even fragment into multiple fronts under humid conditions. This behavior confirms that moisture does not merely act as a plasticizing agent but as an active modulator of the energetic landscape governing detrapping and amorphous-phase connectivity.

In parallel, the semiconductive interfaces exert a decisive influence through their chemical symmetry. Symmetric configurations (OSL-LDPE-OSL and ISL-LDPE-ISL) suppress charge accumulation by more than an order of magnitude and delay packet nucleation, producing broad, low-amplitude profiles with minimal internal field distortion. In contrast, asymmetric configurations, particularly ISL-LDPE-OSL, induce early injection, high-intensity profiles, and steep field gradients, favoring mechanisms associated with failure modes in insulation subjected to high electrical stress. Statistical analysis confirms that these

effects are systematic and reproducible ($p < 0.01$), underscoring the determining role of interfacial symmetry in the dielectric response of the system.

The combination of both experimental axes reveals a unified conceptual framework: moisture modulates volumetric processes, while interfacial symmetry regulates injection fluxes; both factors are coupled and non-additive. Humidified systems with symmetric interfaces exhibit faster transport but moderate charge densities, whereas dry, highly asymmetric systems represent the most critical scenario, with elevated accumulation, severe field distortion, and increased susceptibility to electrothermal aging. This dual-modulation model integrates mechanisms previously treated in isolation and provides a robust approach for understanding the evolution of space charge in modern polymers.

The findings open a generalizable pathway for the advanced design of dielectric materials. Interfacial engineering, without altering the polymer matrix, emerges as an effective strategy to minimize charge accumulation and enhance reliability in HVDC cables and power-electronics insulation structures. Complementarily, environmental control through moderate humidity can be used to stabilize volumetric transport and mitigate internal field distortion. Together, these strategies constitute a scalable approach for designing insulation systems that are more resilient to electrical, thermal, and hygrothermal stresses.

Although the study provides a solid experimental foundation, certain intrinsic constraints should be acknowledged. The moisture content was not quantified through spectroscopic or thermal methods, and the planar parallel-plate geometry does not fully replicate the thermal and mechanical gradients present in extruded cable insulation. In addition, the numerical description adopted here remains phenomenological rather than fully predictive. These limitations are typical of exploratory high-field PEA studies and outline clear directions for future work involving precise moisture quantification, coupled electro-thermal analysis, and validation in cylindrical XLPE-based geometries. Such developments will enable a more general and application-oriented framework for the design of HVDC insulation systems under real operating conditions. Future work incorporating spectroscopic quantification of absorbed moisture (e.g., FTIR, TGA, or sorption analysis) will allow full correlation between water content and trap-depth modification.

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